100 Years
NORTH SYDNEY BOYS HIGH SCHOOL
1912 ~ 2012

MATHEMATICS

2013 Year 8 Assessment Task 1

March 8, 2013

General instructions

- Working time – 50 minutes.
- Commence each new question on a new page.
- Write using blue or black pen. Where diagrams are to be sketched, these may be done in pencil.
- Board approved calculators may be used.
- All necessary working should be shown in every question.
- Attempt **all** questions.
- At the conclusion of the examination, bundle the sheets used in the correct order within this paper and hand to examination supervisors.

Class (please ✔)

- ○ 8MAT.A – Mr Berry
- ○ 8MAT.B – Mr Lowe
- ○ 8MAT.C – Mrs Juhn
- ○ 8MAT.D – Mr Lam
- ○ 8MAT.E – Mr Ireland/Mr Rezcallah

NAME: ....

Marker's use only.

QUESTION	1	2	3	4	**Total**	%
MARKS	/12	/12	/11	/13	/48	

Question 1 (12 Marks) Commence a NEW page. **Marks**

(a) Express these ratios in simplest form:

i. 7 minutes : 1 hour **1**

ii. $\frac{7}{2} : \frac{2}{5}$ **2**

(b) Find the value of a: $\frac{a}{9} = \frac{7}{3}$. **1**

(c) The ratio of radio listeners to TV watchers is 4 : 7. If there are 511 TV watchers, how many radio listeners are there? **2**

(d) Find the simple interest earned if \$7 000 was invested at 5.75% per annum for four years. **2**

(e) Ben and Casey decide to share their weekly pocket money of \$15 in the ratio 5 : 7. How much would each receive? **2**

(f) Divide \$2 248 in the ratio 9 : 3 : 4. **2**

Question 2 (12 Marks) Commence a NEW page. **Marks**

(a) Express $16\frac{2}{3}\%$ as a fraction in simplest form. **2**

(b) A 375 mL can of Coca Cola® contains 33 g of sugar. Assuming 1 mL of Coca Cola® weighs 1 g, what percentage of the drink is sugar? **2**

(c) It is known that watermelons are 92% water. A particular watermelon weighs 3 600 g. How much does the actual water in this watermelon weigh? **1**

(d) Calculate the sale price if a discount of 15% was applied to a purchase of \$27. **2**

(e) Find the commission paid to a real estate agent if they are paid 3.3% of the sale price of a \$600 000 home. **1**

(f) Calculate the percentage profit that Apple makes on an iPhone 5 if the cost of materials + manufacturing is \$207 and the phone retails for \$649. Give your correct to 2 decimal places. **2**

(g) Calculate 23% of 77% of \$45, correct to nearest cent. **2**

Question 3 (11 Marks) Commence a NEW page. **Marks**

(a) Calculate the amount of petrol that can be purchased with \$40.00 at \$1.47 per litre, correct to 2 decimal places. **2**

(b) A school is staffed at a rate of 18.6 students per teacher. How many students would there be at a school with 85 teachers? **2**

(c) A two litre bottle of juice cost \$4.79, whilst a three litre bottle costs \$6.99. Which is the better value? **2**

(d) Convert 120 km/h to metres per second. **2**

(e) Write each expression in the simplest index form:

i. $a^6 \times a^7$. **1**

ii. $\left(x^{11}\right)^3 \div \left(x^2\right)^2$. **2**

Examination continues overleaf. . .

Question 4 (13 Marks) Commence a NEW page. **Marks**

(a) A bottle full of cordial has the 1 : 5 as the ratio of concentrate to water. Another bottle, which has half the capacity of the first bottle (but also full), has a ratio of 1 : 4 for concentrate to water.

The contents of both bottles are tipped into a third container.

i. Draw a diagram indicating the proportion of water and concentrate in the first bottle. **1**

ii. Find ratio of concentrate to water in this third container. **2**

(b) *Inflation* is the phenomenon where prices of good and services increase due to increased costs to produce. It is measured as a percentage of last year's prices.

Assume that inflation in Australia is 2.5% per annum and does not change, and the current (2013) cost of a Big Mac is \$4.95.

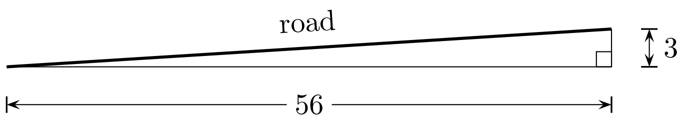
i. How much would the Big Mac cost in 2014 (correct to nearest cent), given the above information? **1**

ii. How much would the Big Mac cost in 2018 (correct to nearest cent)? **1**

iii. In how many years (to the nearest year) will a Big Mac cost \$10.00 due to inflation? **1**

(c) The *gradient* of a road is a measure of its steepness as a ratio or percentage.

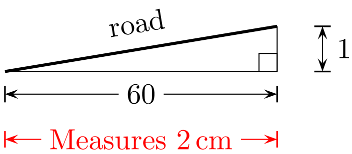
i. A road has a gradient of 3 in 56 (rises 3 metres for every 56 metres of horizontal distance, or $\frac{3}{56}$). **1**



What is the percentage of the steepness, correct to two decimal places?

ii. If the steepness is 5.25%, how many metres would the road rise if the horizontal length was 800 m? **3**

iii. A section of a road that has a steepness of 1 in 60 is shown on a cross sectional map with scale 1 : 250 000. If it measures 2 cm horizontally on the map, how far does it rise in reality, correct to 1 decimal place? **3**



End of paper.

Suggested Solutions

Question 1

(a) i. (1 mark)

$$7\,\text{min} : 1\,\text{h} = 7 : 60$$

ii. (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer.

$$\underbrace{\frac{7}{2}}_{\times 5} : \underbrace{\frac{2}{5}}_{\times 2} = \frac{35}{10} : \frac{4}{10}$$
$$= 35 : 4$$

(b) (1 mark)

$$\frac{a}{9} = \frac{7}{3}$$
$$a = \frac{7}{\cancel{3}} \times \cancel{9}^{3} = 21$$

(c) (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer.

$$R : T = 4 : 7$$
$$R : 511 = 4 : 7$$
$$\underset{\times 511}{\frac{R}{511}} = \underset{\times 511}{\frac{4}{7}}$$
$$R = 292$$

(d) (2 marks)

✓ [0] for entire part if simple interest formula incorrect.
✓ [1] for progress towards answer.
✓ [1] for final answer.

$$\begin{aligned} I &= Prn \\ &= 7\,000 \times 0.0575 \times 4 \\ &= 1\,610 \end{aligned}$$

(e) (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer (penalise here for \$ sign?)

- 5 : 7 implies 12 parts total.
- 1 part is $\frac{15}{12} = \frac{5}{4} = 1.25$.
- $\$1.25 \times 5 = \$6.25$.
- $\$1.25 \times 7 = \$8.75$.
- Hence ratio is \$6.25 : \$8.75.

(f) (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer

- 9 : 3 : 4 implies 16 parts total.
- 1 part is $\frac{2\,248}{16} = 140.5$.
- $140.5 \times 9 = 1\,264.5$.
- $140.5 \times 3 = 421.5$.
- $140.5 \times 4 = 562$.
- Hence ratio is \$1 264.5 : \$421.5 : \$562.

Question 2

(a) (2 marks)

✓ [−1] for each error.

$$\begin{aligned} 16\frac{2}{3}\% &= \frac{16}{100} + \frac{2}{3} \times \frac{1}{100} \\ &= \frac{16}{100}{}^{\times 3}_{\times 3} + \frac{2}{300} \\ &= \frac{48+2}{300} = \frac{50}{300} = \frac{1}{6} \end{aligned}$$

(b) (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer.

$$\frac{33\,\mathrm{g}}{375\,\mathrm{g}} = 8.8\%$$

(c) (1 mark)

$$3\,600\,\mathrm{g} \times 0.92 = 3\,312\,\mathrm{g}$$

(d) (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer.

$$\$27 \times 0.85 = \$22.95$$

(e) (1 mark)

$$600\,000 \times 0.033 = \$19\,800$$

(f) (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer.

$$\frac{\$649}{\$207} = 3.1353\cdots$$

Hence percentage profit is 213.53% (2 d.p.)

(g) (2 marks)

✓ [1] for progress towards answer.
✓ [1] for final answer. (Penalise for rounding?)

$$0.23 \times 0.77 \times \$45 = \$7.97$$

Question 3

(a) (2 marks)

✓ [1] for progress towards answer.

✓ [1] for final answer. (Penalise for units?)

$$\frac{\$40}{\$1.47/\text{L}} = 27.21\,\text{L}$$

(b) (2 marks)

✓ [1] for progress towards answer.

✓ [1] for final answer.

$$\frac{18.6\,\text{students}}{\underset{\times 85}{1\,\text{teacher}}} = \frac{x\,\text{students}}{\underset{\times 85}{85\,\text{teachers}}}$$

$$x = 85 \times 18.6 = 1\,581$$

(c) (2 marks)

✓ [1] for correct method for finding unit pricing.

✓ [1] for final answer.

- $\dfrac{\$4.79}{2} = \$2.395/\text{L}$
- $\dfrac{\$6.99}{3} = \$2.33/\text{L}$
- Second bottle is cheaper.

(d) (2 marks)

✓ [1] for significant progress towards answer.

✓ [1] for final answer.

$$\begin{aligned} 120\,\text{km/h} = \frac{120\,\text{km}}{1\,\text{h}} &= \frac{120\,\text{km}}{60\,\text{min}} \\ &= \frac{120\,000\,\text{m}}{60 \times 60\ \text{sec}} \\ &= 33\frac{1}{3}\ \text{m/s} \end{aligned}$$

(e) i. (1 mark)

$$a^6 \times a^7 = a^{13}$$

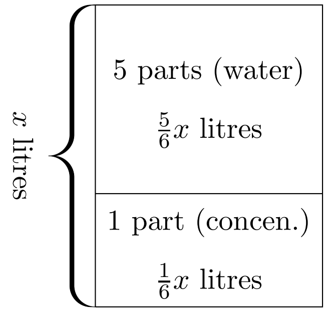
ii. (2 marks)

$$\begin{aligned} \left(x^{11}\right)^3 \div \left(x^2\right)^2 &= x^{33} \div x^4 \\ &= x^{29} \end{aligned}$$

Question 4

(a) i. (1 mark)

Assume x litres. Students can assume 1 L, 100 L etc.



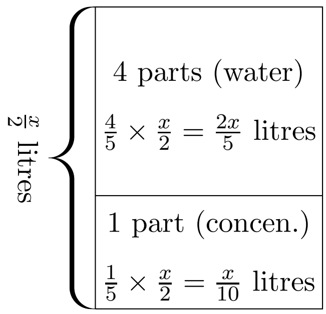
$C : W = 1 : 5$

ii. (2 marks)

✓ [1] for correct diagram (or other significant progress towards answer).

✓ [1] for final answer.

Second bottle:



$C : W = 1 : 4$

- When the two bottles are poured into third container:
  - Concentrate content:

    $$\frac{1}{6}x + \frac{1}{10}x = \frac{8}{30}x \text{ litres}$$

  - Water content:

    $$\frac{5}{6}x + \frac{2}{5}x = \frac{37}{30}x \text{ litres}$$

- Hence $C : W$ in the new mixture is $8 : 37$.

(b) i. (1 mark)

$$\$4.95 \times 1.025 = \$5.07$$

ii. (1 mark)

- NB. Students need not use compound interest formula to obtain mark – can repeatedly multiply by 1.025.

$$\begin{aligned} A &= P(1+r)^n \\ &= \$4.95(1.025)^5 \\ &= \$5.60 \end{aligned}$$

iii. (1 mark)

$$A_{28} = \$4.95(1.025)^{28} = \$9.88$$
$$A_{29} = \$4.95(1.025)^{29} = \$10.13$$

Accept 28 or 29 years (by trial and error to obtain $A \approx \$10$.

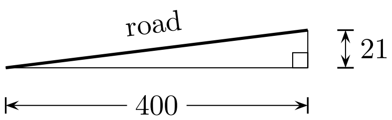
(c) i. (1 mark)

$$\frac{3}{56} = 0.05357\cdots = 5.36\%$$

ii. (3 marks)

✓ [1] for correctly resolving % into fraction.
✓ [1] for $m = \frac{21}{400}$.
✓ [1] for final answer.

$$\begin{aligned} 5.25\% &= \frac{5.25}{100} \\ &= \frac{525}{10\,000}\,{}^{\div 25}_{\div 25} \\ &= \frac{21}{400} \end{aligned}$$



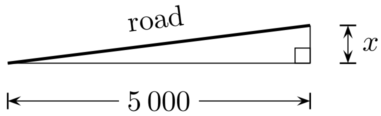
$$\frac{21}{400} = \frac{x}{800}$$
$$\therefore x = 42$$

Hence road rises 42 m for the 800 m horizontal length.

iii. (3 marks)

✓ [1] for 2 cm × 250 000 to obtain real horizontal distance.
✓ [1] for setting up equation.
✓ [1] for final answer.

$$2\,\text{cm} \times 250\,000 = 5\,000\,\text{m}$$



$$\underset{\times 5\,000}{\frac{x}{5\,000}} = \underset{\times 5\,000}{\frac{1}{60}}$$
$$x = \frac{5\,000}{60} = 83.3\,\text{m}$$